SENG 4660 Sprint 2 Review Summary

Sprint Period: February 7 – March 14, 2026

Sprint Goal: Complete the resource import and the construction of basic levels, and achieve full control of playable characters and their collision interaction with the map.

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1. Sprint Goal and Planned User Stories

The goal of this Sprint is to officially enter the game development stage. Based on the project framework established in Sprint 1, the necessary resources will be imported, a basic level (Tilemap) will be set up, and full control of the player character (movement, jumping) and collision detection between the character and the map will be implemented. The planned user stories to be completed are as follows:

- US-01: Player Movement (Left and Right Movement + Jumping)
- US-02: Character and Platform Collision
- Resource Import and Tilemap Basic Setup (Technical Task, Preparatory Work for US-01 and US-02)

2. Completed Work

During the Sprint, the team accomplished the following tasks:

- **Resource Import and Management:**
  - Import the free 2D character sprites and Tilemap tile sets (obtain them from the Unity Asset Store or Kenney.nl).
  - Organize the folder structure of the project (Scenes, Scripts, Sprites, Prefabs, Tilesets, etc.).
- **Tilemap Level Construction:**
  - Created a test Tilemap and drew a simple level layout consisting of ground, platforms and walls.
  - Set up the Tilemap Collider 2D to ensure that the terrain has the correct collision boundaries.
- **Player character control:**
  - Create a player GameObject, add Rigidbody2D, BoxCollider2D and a custom C# script (PlayerController.cs).
  - Implement left-right movement (horizontal speed control) and jumping (applying upward force and detecting the ground) based on the Unity Input System.
  - Adjust the physical parameters (gravity scale, jumping force, movement acceleration) to ensure gameplay is consistent with platform game expectations.
- **Collision detection:**
  - The player collides correctly with the Tilemap terrain and will not pass through the ground or walls.
  - Implement ground detection (using Physics2D.OverlapCircle to check if it is on the ground), and ensure that jumping can only occur on the ground.
  - The test focused on observing the character's behaviors, including

walking, jumping, falling, and collisions. The collision jitter issue was also resolved.
- **GitHub Projects Board Settings and Task Tracking:**
  - Create a Projects board (Board layout) in the GitHub repository, including three columns: "To Do", "In Progress", and "Done".
  - Create individual Issues for each user story (US-01 to US-08) in the product backlog and add them to the "To Do" column of the Kanban board.
  - Add the tasks of this Sprint plan (US-01, US-02 and related technical tasks) to the board, and during the development process, move the corresponding cards from "To Do" to "In Progress" and "Done".
- **Git Version Control:**
  - All newly added resources and codes are submitted through the "feature" branch and undergo code review before being merged into the "main" branch.
  - The submitted information is clear and records the completion status of each function.

3. Sprint Demonstration

At the Sprint review meeting, the team presented the following content to the instructor and classmates:

- At the Sprint review meeting, the team presented the following content to the instructor and classmates:
- In the Unity editor, the project folder structure and imported resources are displayed.
- The Tilemap layout of the test level, showing different terrain blocks (grassland, bricks, platforms).
- The player character can freely move, jump, pass through gaps, and stand on the edge of platforms within the level.
- The detection range on the ground is displayed in the Debug interface, and the detection accuracy is confirmed.

4. Feedback Received

- The feedback collected during the review process includes:
- The character's movement speed feels a bit too fast. It is recommended to adjust the parameters to make it more suitable for novice players.
- The position of the collision point detected on the ground can be adjusted to avoid inaccurate detection of the character on the slope (currently, there is no slope, but one may be added in the future).
- The tile selection in the Tilemap is excellent, and the visual style is clear. However, some decorative elements (such as flowers and grass) could be added to make the levels more lively.
- Currently, there is no sound effect. The jumping and landing movements are a bit monotonous. In the future, we can consider adding basic sound effects.

5. Adjustments and Next Steps

Based on the feedback, the team has decided to do the following in the next Sprint:

- Optimize the player control parameters, adjust the movement speed and jumping force to make it more comfortable.

- Add some decorative tiles to the Tilemap to enhance its visual appeal.
- Start implementing the item collection (US-04) and HUD display of scores (US-03).
- Start designing the first trap that incorporates pranks (the initial attempt for US-08), such as a hidden falling platform.
- Continue to improve the GitHub Projects board, move the new task cards to the corresponding columns, and ensure that all user stories are associated with the board.